

# Ah-Hyeon Lee

M.S. Candidate in IT Engineering, Sookmyung Women's University, Seoul, Korea

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## EDUCATION

**Sookmyung Women's University, Seoul, Korea** (Sep. 2024 – Aug. 2026)

*M.S. in IT Engineering*

- Nominated for Outstanding Thesis Award
- Advisor: Prof. Byung-Gyu Kim (IVPL)
- GPA 4.15/4.3

**Sookmyung Women's University, Seoul, Korea** (Mar. 2019 – Aug. 2024)

*B.S. in IT Engineering*

- GPA 3.29/4.3 (Major 3.37/4.3)

## RESEARCH EXPERIENCE

Graduate Researcher, IVPL, Sookmyung Women's University (Sep. 2024 – Present)

- RAPID: data-adaptive GMM prior + step-adaptive noise — 84% fewer steps, 6.3× faster (40 NFE, FID 5.40 on ImageNet-1K)
- CAUSE: vision–audio–text cross-attention for utterance-level speaker–listener empathy generation
- GPU Server Team Lead: full-stack training infra (hardware–BIOS–OS–library), I/O & thermal optimization

Undergraduate Research Intern, IVPL, Sookmyung Women's University (Dec. 2023 – Aug. 2024)

- Empathy research project: empathy-process modeling for a 3D empathetic avatar service

## PUBLICATIONS (\* corresponding author)

Under review / in preparation

- A. H. Lee, B. G. Kim\*, "RAPID: Robust Adaptive Prior Integration for Diffusion," Pattern Recognition (under review)
- A. H. Lee, Y. J. Cho, B. G. Kim\*, "CAUSE: Sequential Empathetic Listener Reaction Generation from Response Speech," AAAI 2027 (under review)

First author

- A. H. Lee et al., "Adaptive Prior Diffusion: Dataset-Aware GMM Prior for High-Fidelity Image Generation," KMMS Spring 2026 [Best Paper Awards]
- A. H. Lee et al., "An Optimized Diffusion Model Based on Adaptive Prior Distribution," KMMS Autumn 2025 [Best Paper Awards]
- A. H. Lee, B. G. Kim\*, "Attention Module Minification by Low-rank Factorization," KMMS Spring 2025 [Best Paper Awards], and 2 others

Second author

- Y. J. Cho, A. H. Lee et al., "RMSF-ViT: Randomized Multi-Scale Fusion Vision Transformer," CCIS (Springer), MITA 2025 [Best Paper Awards]
- Q. U. A. Aisha, A. H. Lee, B. G. Kim\*, J. Kang, "Emotion Recognition Technique: A Comprehensive Review," Journal of Sciences and Innovation for Sustainable Peace, vol. 3, no. 1, pp. 35–46, Apr. 2025, and 2 others

## FUNDED RESEARCH PROJECTS

Vision-based Emotion Recognition & Empathy Modeling

- Principal Investigator | WISET | Apr. – Oct. 2025

Visual Signal Analysis & Empathy Modeling for User Emotion Recognition

- Principal Investigator | IITP | Sep. – Nov. 2024

Empathetic AI Women Engineering Education (BK21 FOUR, Phase 4)

- Participant | MOE | Mar. 2024 – Aug. 2026, and 2 others

## SKILLS

- Areas: generative models, efficient AI, multimodal fusion, affective computing
- Programming: Python (PyTorch), C++ | Tools: OpenCV, Dlib, CUDA | GPU server management

## **LANGUAGES**

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OPIc IM2 (Mar. 2026)

## **ACTIVITIES**

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Destiny, Rock Band, Sookmyung Women's University, Korea (Mar. 2020 – May. 2023)